

Keynote

Structure, Tradition and Possibility

Theodor Holm Nelson

Leverhulme Visiting Professor

School of Computer Science and Information Technology

University of Nottingham, U.K.

ted@xanadu.net

Science is supposedly about reality, not about tradition, conventions or constructs. Yet computer science seems to me wrongly centered around two traditional, conventional constructs: the simulation of hierarchy and the simulation of paper.

It is a popular myth that “structure” means hierarchy; and it is a popular conception that electronic documents should simulate paper. These two concepts have the additional advantage of being easy to explain to beginners. Accordingly, since the nineteen-forties we have simulated hierarchies to organize computer files, and since the nineteen-sixties we have progressively simulated paper — from “text editing” to “word processing” to “desktop publishing” to the Web (which added one-way links to simulated sheets of paper). Now, merging hierarchy simulation with paper simulation, we have been given Adobe Acrobat (simultaneously simulating hierarchy and paper side by side) and XML (a system for transforming paper simulation into hierarchy and vice versa).

I see these as ideological exercises in completing the hierarchy and paper paradigms, bypassing the vital issues. Rather than imitating the shortcomings of the real world, we should be correcting the insufficiencies of hierarchy and the deficiencies of paper. Things’ being simple-minded and easy to explain does not make them sensible or right.

Hierarchy is wrong and insufficiently general. Vital forms of information structure cannot be properly represented by hierarchy — such structures as parallelism, cross-connection, interpenetration and polypresence (one item in many places). In designing your file arrangements, one set of relations must be force-chosen and artificially made central. You must select one and only one hierarchy — such as time, project, importance — at the cost of representing others. This gives you base addresses that you must then overlay by such other conventions as aliases, shortcuts and database.

Paper simulation is wrong and insufficiently general. Vital document structures and issues are ignored in today’s paper-simulating formats — overlay and document parallelism, of which annotation and controversy swirl are key examples; issues of version management, quotability with managed rights, availability of original context of quotations, and structures of arbitrary connection.

Setting these traditional structures aside opens many possible alternatives that have not been explored. As existence proofs of wholly different structures I offer two minimalist examples.

As a more generalized structure than hierarchy, I recommend zzstructure. In zzstructure we build data complexes out of multidimensional elastic blocks (equivalent to general graph structure, but with row-and-column views and operations). This allows us to build criss-crossing lists where any unit may be on many lists simultaneously. Thus all desired relations may be represented at once without conflict or cross-reference. This makes it easy to create rich new environments of linked personal information, new forms of programming and new forms of interactive graphics.

As a generalization and improvement of paper, I offer a parallel document format: VLIT™ — a Virtual LITerary system, a layered structure for ever-quotable, ever-changing hypertext (a fresh example of the Xanadu® model). A VLIT document is distributed as a list of addresses of quoted material and optional overlays for organizing, decorating and linking. Like the other Xanadu designs, it offers a variety of nonhierarchical structures and arrangements with many alternative views (WYSIWYC, What You See Is What You Choose). Structure and links are applicative, so that alternative structures may be turned on and off and links seen or not. Links may of course be followed in any direction. Quotations come from the rightsholder and the original context remains available; content does not expire and links do not break. VLIT documents are set up under the assumption of constant revision, managing versions automatically and keeping them available. Incremental changes, even of published documents, leave all relationships valid.

Demonstration examples of such systems have been around for some time. Now it is time to build user worlds on these principles. I am certain that such worlds can be made more sensible and habitable for ordinary people, and more suited to the concerns of ordinary people, than the conventional, traditional structures we have been building all this time.

In the conventional paradigm, users must concern themselves with file naming, hierarchy and location; this tradition may be dropped. In the conventional paradigm, we must distinguish between “applications” (walled-off zones of incompatible function allowing one level of undo) from the operating system (dangerous operations, no undo). These unnecessary traditions may be dropped. In the conventional paradigm, users need many different “applications” — such as “database,” “word processing,” “spreadsheet” — to manage their information worlds. These traditions may now be dropped.

Fifty years of computer tradition need to be unwound and undone to open the possibilities we have even ceased to suspect. Trans© T. Nelson.